

- 6.1** When an interrupt occurs, an operating system
- (a) ignores the interrupt
 - (b) always changes state of interrupted process after processing the interrupt
 - (c) always resumes execution of interrupted process after processing the interrupt
 - (d) may change state of interrupted process to 'blocked' and schedule another process

[1997 : 1 M]

- 6.2** A user level process in Unix traps the signal sent on a Ctrl-C input, and has a signal handling routine that saves appropriate files before terminating the process. When a Ctrl-C input is given to this process, what is the mode in which the signal handling routine executes?

- (a) kernel mode
- (b) super user mode
- (c) privileged mode
- (d) user mode

[2005 : 1 M]

- 6.3** A student wishes to create symbolic links in a computer system running Unix. Three text files named "file1", "file2" and "file3" exist in her current working directory, and the student has read and write permissions for all three files. Assume that file1 contains information about her hobbies, file2 contains information about her friends and file3 contains information about her courses. The student executes the following sequence of commands from her current working directory

```
ln -s file1 file2
```

```
ln -s file2 file3
```

Which of the following types of information would be lost from her file system?

- I. Hobbies
- II. Friends
- III. Courses

- (a) I and II only
- (b) II and III only
- (c) II only
- (d) I and III only

[2005 : 1 M]

- 6.4** The shell command

```
find -name password -print
```

is executed in /etc directory of a computer system running Unix. Which of the following shell commands will give the same information as the above command when executed in the same directory?

- (a) ls passwd
- (b) cat passwd
- (c) grep name passwd
- (d) grep print passwd

[2005 : 1 M]

- 6.5** Consider n jobs $J_1, J_2, \dots, J_n$ such that job J_i has execution time t_i and a non-negative integer weight w_i . The weighted mean completion time

of the jobs is defined to be $\frac{\sum_{i=1}^n w_i T_i}{\sum_{i=1}^n w_i}$ where T_i is

the completion time of job J_i . Assuming that there is only one processor available, in what order must the jobs be executed in order to minimize the weighted mean completion time of the jobs?

- (a) Non-decreasing order of t_i
- (b) Non-increasing order of w_i
- (c) Non-increasing order of $w_i t_i$
- (d) Non-increasing order of w_i/t_i

[2007 : 2 M]

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Answers Miscellaneous

6.1 (d) 6.2 (a) 6.3 (b) 6.4 (a) 6.5 (d)

Explanations Miscellaneous**6.1 (d)**

An interrupt is a signal from a device attached to a computer or from a program within the computer that causes the main program that operates the computer to stop and figure out what to do next. After the interrupt signal is sensed, it may change state of interrupted process to 'blocked' and schedule another process.

6.2 (a)

When user level process trapping the Ctrl+C signal then the trap signal is going through system call and that's why mode changed to Kernel mode from user mode and then the request is handling.

One more thing kernel mode and privilege mode are same, answer is Kernel mode (privilege mode).

6.3 (b)

In unix system: \$ ln -s file 1, file 2 command replace existing file 2 by file 1. If file 2 exist then it is first removed then replace with file 1.

ln -s file 1 file 2

ln -s file 2 file 3

So, file 2 replace file 3 the file replace file 2 i.e. friends and courses are lost and hobbies are remaining.

6.5 (d)

To minimize the weighted mean completion time, jobs must be executed in increasing order of weight/time like FCFS scheduling algorithm. If we go with decreasing order of weight/time then it suffers from convoy effect.

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